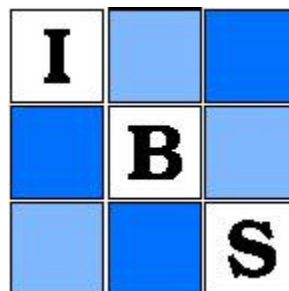


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Author(s): Stephen J. Senn, Richard A. Brown and K. E. James

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READER REACTION

Estimating Treatment Effects in Clinical Trials Subject to Regression to the Mean

Stephen J. Senn

Department of Mathematics and Computer Studies, Dundee College of Technology,
Bell Street, Dundee DD1 1HG, United Kingdom

and

Richard A. Brown

Department of Mathematics, University of Dundee,
Perth Road, Dundee DD1 4HN, United Kingdom

1. The Method of Moments

If patients are selected for a clinical trial because they have been measured as having high or low values, as the case may be, of some variable of interest (e.g., blood pressure) and if the correlation between first and subsequent measurements is less than one, then it is to be expected that if as a group they are measured again their average measurement value will have regressed to the mean. This well-known phenomenon of regression to the mean thus presents a problem for the evaluation of uncontrolled clinical trials and, clearly, may cause difficulties in other fields of research as well. In a paper on this subject, James (1973) proposes a method of estimating the regression effect using moments and is able to obtain estimators of a fairly simple form, both for the regression effect and for various parameters of interest, by assuming that the proportion of the population selected for treatment is known.

James (1973) assumes that in the absence of any treatment and in the absence of any selection procedure, first measurements X and second measurements Y will have a bivariate normal distribution with

$$E(X) = E(Y) = \mu,$$

$$\text{var}(X) = \text{var}(Y) = \sigma^2,$$

and the correlation between X and Y , ρ .

He further supposes that individuals with $X > k$ are selected for treatment and that the effect of the treatment is such that the second measurement, Y_t (where the subscript denotes that a treatment has taken place), is modelled by

$$Y_t - \mu = \gamma\rho(X - \mu) + \varepsilon,$$

where ε is distributed normally and independently of X with mean 0 and variance $\sigma^2(1 - \rho^2)$, and γ is a treatment parameter. The case $\gamma < 1$ corresponds to an intervention which succeeds in closing the gap between the selected “abnormal” population and the population mean.

By using the method of moments, James (1973) obtains the following estimators:

$$\begin{aligned}\hat{\mu} &= \bar{x} - a\hat{\sigma}, \\ \hat{\sigma}^2 &= S_x^2/[a(z - a) + 1], \\ \hat{\rho} &= \{\hat{\beta}^2[a(z - a) + 1] - S_y^2/\hat{\sigma}^2 + 1\}^{1/2}, \\ \hat{\gamma} &= \hat{\beta}/\hat{\rho},\end{aligned}$$

where $\hat{\beta}$ is the estimated slope for Y_i on X ; $\bar{x}$ is the mean and S_x^2 is the variance for the truncated sample of X values; S_y^2 is the variance of the corresponding sample of Y_i values with

$$z = (k - \mu)/\sigma \quad \text{and} \quad a = \phi(z)/[1 - \Phi(z)];$$

and where $\phi(z)$ and $\Phi(z)$ are the standard normal density and distribution functions, respectively.

To obtain formulas for the variances of these estimators, however, a number of oversights are made. These will be outlined below.

2. Variances for the Method of Moments

According to James (1973), the variance of $\hat{\sigma}^2$ is given by

$$\text{var}(\hat{\sigma}^2) = 2\sigma_x^4/\{[a(z - a) + 1]^2(n - 1)\},$$

where σ_x^2 is the variance of the truncated X population. Comparing with the expression for $\hat{\sigma}^2$, we see that this will indeed be the variance of σ^2 if z is known and fixed, and the variance of S_x^2 is $2\sigma_x^4/(n - 1)$. However, the X values, being members of a truncated normal population, are not normally distributed and therefore the variance of S_x^2 is not equal to $2\sigma_x^4/(n - 1)$.

Next, James (1973) obtains an expression for the variance of $\hat{\sigma}$,

$$\text{var}(\hat{\sigma}) \approx \sigma_x^4/\{2\sigma^2[a(z - a) + 1]^2(n - 1)\},$$

using a Taylor's approximation which leads to an expression for the variance of $\hat{\mu}$:

$$\text{var}(\hat{\mu}) \approx \sigma_x^2/n + a^2\sigma_x^4/\{2\sigma^2[a(z - a) + 1]^2(n - 1)\}.$$

This expression ignores the covariance between $\bar{x}$ and S_x^2 . Of course, as is well known, $\bar{x}$ and S_x^2 are independent for a sample drawn from a normal population. Here, however, the population is that of a truncated normal.

Similar problems arise in the derivation of formulas for the variances of $\hat{\rho}$ and $\hat{\gamma}$ in that some covariance terms are not allowed for. Furthermore, in these derivations the variance of $\hat{\beta}$ is assumed to be equal to $\sigma^2/\sum (x - \bar{x})^2$. A similar formula is applicable in the context of linear regression where, however, σ^2 is taken to be the conditional variance of Y given X , whereas in the paper by James (1973), σ^2 is the variance of the untruncated population of X .

As an example of the consequences of these oversights, consider the expression for the variance of $\hat{\sigma}^2$. Since

$$\sigma_x^2 = \sigma^2[a(z - a) + 1],$$

this reduces to $\text{var}(\hat{\sigma}^2) = 2\sigma^4/(n - 1)$, implying that the choice of truncation point has no effect on the variance of the estimator. A corrected expression for the variance is

$$V/[a(z - a) + 1]^2,$$

where V is the variance of S_x^2 and

$$V = \sigma^4 \{n[2 + (z + z^3)a - 5z^2a^2 + 8za^3 - 4a^4] \\ + [(3z - z^3)a - (4 - 7z^2)a^2 - 12za^3 + 6a^4]\}/[n(n - 1)].$$

A corrected expression for the variance of $\hat{\mu}$ is

$$\text{var}(\hat{\mu}) = \sigma_x^2/n + a^2V/\{4\sigma^2[a(z - a) + 1]^2\} \\ - \sigma^2a[(z^2 - 1)a - 3za^2 + 2a^3]/\{[a(z - a) + 1]n\}.$$

3. The Method of Maximum Likelihood

An approach using maximum likelihood and which does not require knowledge of the truncation proportion is outlined below. The opportunity will also be taken to generalise James's problem in order to deal with various kinds of sampling associated with the bivariate normal.

Cohen (1955) describes four kinds of sample which may be taken in connection with bivariate normal distributions: truncated, censored, selected, and complete samples. For a *truncated* sample, the sampling procedure is continued until n specimens for which $X \geq k$ are collected. It is not possible to observe values for which $X < k$ and the number of eliminated X values is unknown. Subsequent Y measurements are made on the n chosen individuals. In the case of a *censored* sample, both X and Y are recorded if $X \geq k$. However, a count is kept of censored specimens for which $X < k$, although neither the X nor Y values of such specimens are recorded. For *selected* samples, full measurement is made and recorded for X for all values whether or not $X \geq k$. However, only Y values corresponding to $X \geq k$ are recorded. Finally, a *complete* sample is one for which X and Y are measured whether or not $X \geq k$.

Maximum likelihood estimators to be used with all four types of sample are derived by Cohen (1955). His estimators cannot be used here, however, for two reasons. First, he does not assume that $E(X) = E(Y)$ nor that $\text{var}(X) = \text{var}(Y)$. From the distributional point of view, this means that the regression to the mean problem is simply a special case of Cohen's problem. The effect of incorporating the conditions $E(X) = E(Y)$ and $\text{var}(X) = \text{var}(Y)$, however, is to impose constraints on a previously unconstrained problem with the result that the optimal points may be altered. The second reason is that Cohen does not allow for a treatment effect.

Following Cohen (1955), one can deal with the problems of truncation, censoring, and selection by writing the likelihood function in the form

$$L = G \times \exp[-\frac{1}{2} \sum (x_i - \mu)^2 / \sigma^2] / (2\pi\sigma^2)^{n/2} \\ \times \exp\{-\frac{1}{2} \sum [(y_i - \mu) - \gamma\rho(x_i - \mu)]^2 / [\sigma^2(1 - \rho^2)]\} / [2\pi\sigma^2(1 - \rho^2)]^{n/2},$$

where x_i and y_i are sample values for the treatment group—that is to say, first and second measurement values for individuals chosen for treatment on the basis of their first measurement—and G is a restriction function depending on the type of sample.

The following restriction functions are appropriate:

Truncated sample

$$G = \{1/[1 - \Phi(z)]\}^n$$

Censored sample

$$G = [\Phi(z)]^m$$

Selected sample

$$G = [1/(2\pi\sigma^2)]^{m/2} \exp[-\frac{1}{2} \sum (x_c - \mu)^2/\sigma^2]$$

Complete sample

$$G = [1/(2\pi\sigma^2)]^{m/2} \exp[-\frac{1}{2} \sum (x_c - \mu)^2/\sigma^2] \\ \times 1/[2\pi\sigma^2(1 - \rho^2)]^{m/2} \exp\{-\frac{1}{2} \sum [y_c - \rho x_c + (1 - \rho)\mu]^2/[\sigma^2(1 - \rho^2)]\}$$

In these expressions, x_c and y_c are sample values for individuals rejected for treatment on the basis of their initial measurement values and m is the number of individuals in the below truncation group.

The maximum likelihood solution for any given type of problem can then be obtained by standard numerical methods.

4. An Example

Davis (1976) reports a study undertaken by the Lipid Research Clinics, in which men aged 35 to 39 years were screened for cholesterol levels. Those with levels in excess of 265 mg% proceeded further in the trial and prior to any treatment had their cholesterol levels tested again. The following results were obtained, in terms of log cholesterol:

$$\begin{aligned} \bar{X} &= 5.676 & \bar{Y} &= 5.634 \\ S_x^2 &= 0.00728 & S_y^2 &= 0.01348 \\ \hat{\beta} &= 0.792 \end{aligned}$$

The truncation point in log cholesterol was 5.580 and was believed to correspond to the 95th percentile on the basis of prior information. By applying James's (1973) method, Davis obtained the following results:

$$\hat{\mu} = 5.207, \quad \hat{\sigma} = 0.228, \quad \hat{\rho} = 0.91.$$

These estimates can be used as initial values for maximum likelihood estimation using the derivatives of the likelihood function. An initial value of $\hat{\gamma} = 1.0$ is assumed. After several iterations the following estimates are produced:

$$\begin{aligned} \hat{\mu} &= 5.390 \quad (2.71/n) \\ \hat{\sigma} &= 0.186 \quad (0.196/n) \\ \hat{\gamma} &= 0.985 \quad (0.271/n) \\ \hat{\rho} &= 0.861 \quad (0.538/n) \end{aligned}$$

The figures in parentheses are variances of the estimates where n is the number of subjects proceeding to the second cholesterol measurement. Even if the figure of 3484 persons quoted by Davis represents the number presenting for second measurement, the figure of 0.985 for $\hat{\gamma}$ is less than two standard errors from 1, the value it should have were no treatment effect present.

Davis (1976) also reports external estimates obtained from an untruncated sample of 1346 individuals which yielded values of

$$\hat{\mu} = 5.331, \quad \hat{\sigma} = 0.170, \quad \hat{\rho} = 0.82.$$

It is noticeable that the results obtained by maximum likelihood on the truncated sample are much closer to these values than those obtained using James's (1973) method of moments.

5. Conclusion

In view of the drawbacks already noted to James's (1973) method of moments, it would seem that there is little advantage in using this approach and that maximum likelihood is preferable.

Although no investigation of robustness has been carried out here, it seems reasonable to suppose that neither the method of moments nor the method of maximum likelihood will prove particularly robust with respect to departure from normality, since, presumably, "tail" area peculiarities will have a considerable effect on the shape of a truncated distribution. The fact, therefore, that some estimation of the regression effect is possible for uncontrolled clinical trials should not, as James (1973) notes, provide an excuse for doing without controls.

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RESPONSE

Senn and Brown are quite correct in pointing out that normality cannot be assumed for the truncated distribution when deriving variance formulas for the $\hat{\mu}$, $\hat{\sigma}$, $\hat{\rho}$, and $\hat{\gamma}$. The maximum likelihood approach does have appeal and may be more robust. It is certainly appropriate that this avenue would be explored; however, before firm conclusions can be drawn, several points should be considered:

- (i) Variance estimates of $\hat{\mu}$, $\hat{\sigma}$, $\hat{\rho}$, and $\hat{\gamma}$ can be derived without the assumption of normality. Indeed, Senn and Brown provide the formula for the variance of $\hat{\sigma}^2$. Although complex, one would expect that the computation required for these formulas would be less than the maximum likelihood approach, which requires an iterative procedure. The method of moments estimates can be calculated by hand, whereas a computer would usually be required for maximum likelihood estimation. Senn and Brown do not address the difference in computational cost, nor do they discuss a stopping rule for the iterative procedure.
- (ii) While the maximum likelihood approach provided somewhat better estimates for the Lipid Research Clinic data, this, of course, is only a sample of one case and it may well be prudent to examine the two estimation approaches over a number of cases with varying truncation points. This could be accomplished quite easily with simulation experiments.
- (iii) Senn and Brown use the method of moments estimates as input into their maximum likelihood procedures (with the exception of gamma, which was set equal to one). Since they do not discuss alternative methods of providing these initial estimates, it appears that they are suggesting a two-step procedure in which the initial parameter estimates are obtained by method of moments and then fed into the maximum likelihood procedure for refinement.

- (iv) Although it is recognized that normality does not hold for the truncated distribution, the effect of this nonnormality on the variance estimates is still open to question.

K. E. James

Veterans Administration
Cooperative Studies Program Coordinating Center
3801 Miranda Avenue
Palo Alto, California 94304, U.S.A.